Sahana Hariharan

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Education

University of Illinois at Urbana-Champaign GPA: 3.96/4.0
Master of Computer Science *Aug 2026 – May 2027*
Bachelor of Science in Computer Science & Statistics *Aug 2023 – May 2026*

Relevant Coursework: Algorithms & Models of Computation, Computer Systems, Text Information Systems, Database Systems, Statistical Learning, Machine Learning, Cloud Networking, Software Engineering

Technical Skills

Programming Languages: Python, Java, C/C++, Go, JavaScript, TypeScript, SQL, HTML/CSS, R, Arduino C
Frameworks: Flask, React.js, Node.js, REST APIs, Firebase, MongoDB, Neo4j, AWS, Docker, Databricks
Libraries: Pandas, NumPy, PyTorch, TensorFlow, OpenCV, Scikit-learn, Plotly, Hugging Face Transformers
Developer Tools: Git, GitHub, XCode, Eclipse, VSCode, Atlassian (Jira), Unit Testing, Figma

Experience

Software Engineering Intern | Google (YouTube) May 2026 – Aug 2026
- Refactoring the YouTube v3 Public API for playlist management and migrating legacy monolithic data models (YTPBPlaylists) to a decoupled, microservices-based architecture
- Designing and implementing new recs3 operators within the PlaylistDecorationService using C++ and Google's Boq scaffolding framework to scale support for dynamic playlist types
- Eliminating architectural technical debt by optimizing service-to-service communication to ensure high-throughput, low-latency API performance for millions of developer requests

Data Platform Engineer Intern | Dow Jan 2026 – May 2026
- Developed a pipeline using Azure DevOps to help Dow employees deploy, manage, and delete Databricks Genie Spaces, and maintain their condition from development to production environments
- Implemented automated governance and validation checks, ensuring compliance with internal policies and reducing potential deployment errors

Data Engineering Intern | KPMG Jun 2025 – Aug 2025
- Engineered a low-discrepancy, message-passing Monte Carlo engine that streamlined monetary-unit sampling, improving sample coverage consistency and cutting simulation run-time by 40%
- Developed PySpark pipelines to validate synthetic audit datasets and implemented automated QA checks to ensure robust data integrity across various ERP formats for robust ETL testing
- Implemented a unified Databricks pipeline (within KPMG's proprietary workflow) to extract and format various ERP exports (ex. SAP S/4HANA, NetSuite), streamlining pre-processing and debugging

Software Engineering Intern | Prognosis May 2024 – Aug 2024
- Leveraged Large Language Models (LLMs) to transform 3,000+ unstructured patient records into structured knowledge graphs, enabling AI-driven predictions of diseases and personalized treatment recommendations
- Conducted a comparative analysis of Node2Vec and OpenAI embedding methods within a Neo4j database to evaluate their effectiveness in generating meaningful patient knowledge graphs
- Developed a responsive web application using React.js, integrating a robust front-end with an intuitive user interface to display patient information in a dynamic, searchable table

Projects

Deepfake Detection Model | Disruption Lab Sept 2024 – Jan 2025
- Partnered with an NGO to build a digital content authenticity system supporting global trust and safety efforts
- Trained 15+ deepfake detection models (CNNs, ViTs) on a 20K+ media dataset using PyTorch and TensorFlow
- Developed model testing and flagging pipelines to reduce false positives by 25%, supporting real-time content moderation
- Integrated confidence-based thresholds and auto-flagging logic into backend services using Flask and REST APIs

Simulation-Based Inference Research | Department of Statistics at UIUC Aug 2025 – Dec 2025
- Studied Simulation-Based Inference techniques for Bayesian parameter estimation, including Approximate Bayesian Computation (ABC) and Neural Posterior Estimation (NPE)
- Applied methods to simulated (Lotka–Volterra predator–prey) and real-world financial time series data, analyzing identifiability limits and posterior predictive failures